Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate 2023
Deferred Examinations

Marking Scheme

Biology

Higher Level

Note to teachers and students on the use of published marking schemes

Marking schemes published by the State Examinations Commission are not intended to be standalone documents. They are an essential resource for examiners who receive training in the correct interpretation and application of the scheme. This training involves, among other things, marking samples of student work and discussing the marks awarded, so as to clarify the correct application of the scheme. The work of examiners is subsequently monitored by Advising Examiners to ensure consistent and accurate application of the marking scheme. This process is overseen by the Chief Examiner, usually assisted by a Chief Advising Examiner. The Chief Examiner is the final authority regarding whether or not the marking scheme has been correctly applied to any piece of candidate work.

Marking schemes are working documents. While a draft marking scheme is prepared in advance of the examination, the scheme is not finalised until examiners have applied it to candidates' work and the feedback from all examiners has been collated and considered in light of the full range of responses of candidates, the overall level of difficulty of the examination and the need to maintain consistency in standards from year to year. This published document contains the finalised scheme, as it was applied to all candidates' work.

In the case of marking schemes that include model solutions or answers, it should be noted that these are not intended to be exhaustive. Variations and alternatives may also be acceptable. Examiners must consider all answers on their merits, and will have consulted with their Advising Examiners when in doubt.

Future Marking Schemes

Assumptions about future marking schemes on the basis of past schemes should be avoided. While the underlying assessment principles remain the same, the details of the marking of a particular type of question may change in the context of the contribution of that question to the overall examination in a given year. The Chief Examiner in any given year has the responsibility to determine how best to ensure the fair and accurate assessment of candidates' work and to ensure consistency in the standard of the assessment from year to year. Accordingly, aspects of the structure, detail and application of the marking scheme for a particular examination are subject to change from one year to the next without notice.

Introduction

The marking scheme is a guide to awarding marks to candidates' answers. It is a concise and summarised guide and is constructed so as to minimise its word content. Examiners must conform to this scheme and may not allow marks for answering outside this scheme. The scheme contains key words, terms and phrases for which candidates may be awarded marks. This does not preclude synonyms or terms or phrases which convey the same meaning as the answer in the marking scheme. Although synonyms are generally acceptable, there may be instances where the scheme demands an exact scientific term or unequivocal response and will not accept alternatives. The descriptions, methods and definitions in the scheme are not exhaustive and alternative valid answers are acceptable. If it comes to the attention of an examiner that a candidate has presented a valid answer and there is no provision in the scheme for accepting this answer, then the examiner must first consult with his/ her advising examiner before awarding marks. As a general rule, if in doubt about any answer, examiners should consult their advising examiner before awarding marks.

How to use the marking scheme

- Where only one answer is required alternative answers are separated by '<u>**or**</u>'.
- Where multiple answers are required each word, term or phrase for which marks are allocated is separated by a solidus (/) from the next word, term or phrase.
- The mark awarded for an answer appears in **bold** next to the answer, e.g. **3**.
- Where there are several parts in the answer to a question, the mark awarded for each part appears in brackets, e.g. **5(4)** means that there are five parts to the answer, each part allocated **4 marks**.
- The answers to subsections of a question may not necessarily be allocated a specific mark; e.g. there may be six parts to a question – (a), (b), (c), (d), (e), (f) and a total of **20 marks** allocated to the question. The marking scheme might be as follows, **2(4) + 4(3)**. This means that the first two correct answers encountered are awarded **4 marks** each and each subsequent correct answer is awarded **3 marks**.
- A word or term that appears in brackets ( ) is not a requirement of the answer, but is used to contextualise the answer or may be an alternative valid answer.

Some examples of the marking process

1. **Key words or terms or phrases may be awarded marks, only if presented in the correct context.**

   Sample question: *Outline how water from the soil reaches the leaf.*

   Marking scheme states: Concentration gradient / osmosis / root hair / root pressure / cell to cell / xylem / transpiration or evaporation / cohesion (or explained) or adhesion (or capillarity or explained) or tension (or explained). ***Any six*** **6(3)**

   Sample answer: *Water is drawn up the xylem by osmosis.*

   Although the candidate has presented two key terms (xylem, osmosis), the statement is incorrect and the candidate can only be awarded **3 marks** for referring to the movement of water through the xylem.

2. **Cancelled answers**

   The following is an extract from **S.63o** *Instructions to Examiners, 2021 (for subjects being marked online)* (section 5.4, p.19):

   > "*Where a candidate answers a question or part of a question once only and then cancels the answer, you should ignore the cancelling and treat the answer as if the candidate had not cancelled it.*"

   Sample question: *What is pollination?*

   Marking scheme states: Transfer of pollen / from anther / to stigma. **3(3)**

   Sample answer: ~~*Transfer of pollen by insect to stigma.*~~

   The candidate has cancelled the answer and has not made another attempt to answer the question and may be awarded **2(3)** marks.

   If an answer is cancelled and an alternative version given, the cancellation should be accepted and marks awarded, where merited, for the un-cancelled version only.

   If two (or more) un-cancelled versions of an answer are given to the same question or part of a question, both (or all) should be marked and the answer accepted that yields the greater (greatest) number of marks. Points may not, however, be combined from multiple versions to arrive at a manufactured total.

3. **Surplus answers: [only in Section A] - A surplus wrong answer cancels the marks awarded for a correct answer.**

   (i) **Sample question 1**: *The walls of xylem vessels are reinforced with…………*

   Marking scheme states: Lignin **4 marks**

   Sample answer: *Chitin, lignin*

   There is a surplus incorrect answer, therefore the candidate scores **4 – 4 = 0 marks**.

Sample answer: ~~*Lignin*~~

The answer, which is correct, has been cancelled by the candidate, but there is no additional or surplus answer, therefore the candidate may be awarded **4 marks**.

Sample answer: *Lignin, ~~chitin~~*

There is a surplus answer, which is incorrect, but it has been cancelled and as the candidate has given more than one answer (i.e. the candidate is answering the question more than once only), the cancelling can be accepted and s/he may be awarded **4 marks**.

(ii) **Sample question 2**: *Name the four elements that are always present in protein.*

Marking scheme states: Carbon / hydrogen / oxygen / nitrogen **4(3)**

Sample answer: *Carbon, hydrogen, oxygen, nitrogen, calcium*

There is a surplus answer, which is incorrect, which cancels one of the correct answers, therefore the candidate is awarded **3(3)** marks.

Sample answer: *Carbon, hydrogen, oxygen, calcium*

There is no surplus answer – there are three correct answers, and therefore the candidate is awarded **3(3)** marks.

Sample answer: *Carbon, hydrogen, oxygen, calcium, aluminium*

There is a surplus answer, which is incorrect, and cancels one of the three correct answers, therefore the candidate is awarded **2(3)** marks.

Sample answer: *Carbon, hydrogen, oxygen, calcium, ~~aluminium~~*

There is a surplus answer, which is incorrect, but it has been cancelled so the candidate may be awarded **3(3)** marks.

In the other sections of the paper (Sections B and C), there may be instances where a correct answer is nullified by the addition of an incorrect answer. This happens when the only acceptable answer is a specific word or term. Each such instance is indicated in the scheme by an asterisk *.

Annotations used in the marking

The scripts were marked by examiners using an online marking platform. The following table illustrates the various annotations (symbols) applied by the examiners when marking the scripts. The meaning and use of each of the annotations applied are also explained in the table. These annotations will be seen on a script if viewed as part of the appeal process. Annotations applied by an examiner will be viewed in red. Scripts that were also marked by an advising examiner will show annotations in a green colour.

Annotation	Meaning
	This symbol indicates a correct response / answer.
	This symbol indicates that one mark has been awarded.
	This symbol indicates that two marks have been awarded.
	This symbol indicates an incorrect response /answer.
	Surplus incorrect answer. A surplus incorrect answer has cancelled a correct answer.
	This symbol is placed on all blank pages or part of page to indicate it has been seen by the examiner.
	This symbol can be used by an examiner to indicate a part of a question answer of significance.
	This symbol is used to indicate a correct response for a diagram.
	This symbol is used to indicate an incorrect response for a diagram.
	This symbol is used to indicate a correct response for a label on a diagram.
	This symbol is used to indicate an incorrect response for a label on a diagram.

Bonus marks for answering through the medium of Irish

Bonus marks at the rate of 10% of the marks obtained will be given to a candidate who answers entirely through Irish and who obtains 75% or less of the total mark available in (i.e. 300 marks or less). In calculating the bonus to be applied, decimals are always rounded down, not up ¬ e.g., 4.5 becomes 4; 4.9 becomes 4, etc. See below for when a candidate is awarded more than 300 marks.

Marcanna Breise as ucht freagairt trí Ghaeilge

Léiríonn an tábla thíos an méid marcanna breise ba chóir a bhronnadh ar iarrthóirí a ghnóthaíonn níos mó ná 75% d'iomlán na marcanna.

N.B. Ba chóir marcanna de réir an ghnáthráta a bhronnadh ar iarrthóirí nach ngnóthaíonn níos mó ná 75% d'iomlán na marcanna don scrúdú. Ba chóir freisin an marc bónais sin **a shlánú síos**.

Tábla 400 @ 10%

Bain úsáid as an tábla seo i gcás na n-ábhar a bhfuil 400 marc san iomlán ag gabháil leo agus inarb é 10% gnáthráta an bhónais.

Bain úsáid as an ngnáthráta i gcás 300 marc agus faoina bhun sin. Os cionn an mharc sin, féach an tábla thíos.

Bunmharc	Marc Bónais
301 - 303	29
304 - 306	28
307 - 310	27
311 - 313	26
314 - 316	25
317 - 320	24
321 - 323	23
324 - 326	22
327 - 330	21
331 - 333	20
334 - 336	19
337 - 340	18
341 - 343	17
344 - 346	16
347 - 350	15

Bunmharc	Marc Bónais
351 - 353	14
354 - 356	13
357 - 360	12
361 - 363	11
364 - 366	10
367 - 370	9
371 - 373	8
374 - 376	7
377 - 380	6
381 - 383	5
384 - 386	4
387 - 390	3
391 - 393	2
394 - 396	1
397 - 400	0

Section A	**Best 5**	**100**

Question 1	**Best five answers from (a) – (f)**	**3(6) + 2(1)**
(a)	*Name the **three** elements that are present in **all** fats (lipids).* Carbon (or C) **and** hydrogen (or H) **and** oxygen (or O)	
(b)	*Give **one** difference between fats and oils at room temperature.* Fats are solid **and** oils are liquids	
(c)	*A triglyceride is composed of three molecules of fatty acids attached to another molecule. Name this other molecule.* *Glycerol	
(d)	*Name a test that can be used to demonstrate the presence of fat (lipid) in a food sample.* Brown paper (test) **or** Sudan III (test)	
(e)	*Give **one** metabolic function of fats (lipids) in living organisms.* Energy store	
(f)	*Give **one** structural function of fats (lipids) in living organisms.* Forms cell membrane **or** protective layer around kidney **or** forms adipose tissue **or** other correct	

Question 2		**2(5) + 1(3) + 7(1)**
*In each of the following cases, state **two** features that show how the named structure is adapted for its function.*		
(a)	*The small intestine for the absorption of food.* Highly folded / has villi or has microvilli / has a rich network of blood capillaries / walls are one cell thick / other correct	***Any two***
(b)	*Alveoli for the exchange of gases.* Moist / has a rich network of blood capillaries / walls are one cell thick / elastic / other correct	***Any two***
(c)	*An insect-pollinated flower for attracting insects.* Scented / colourful petals / has nectar / other correct	***Any two***
(d)	*A root for the uptake of water.* Has root hairs / walls are one cell thick / hydrotropism / xylem / other correct	***Any two***
(e)	*The lens for focusing light onto the retina.* Curved / transparent / surrounded by muscle / fluid on both sides / flexible / other correct	***Any two***

Question 3	2(5) + 5(2)
(a)	*Why is it important that carbon is recycled in nature?* So other organisms can use it **<u>or</u>** so it doesn't run out
(b)	*Name processes **A and B** in the diagram.* A: Combustion B: Respiration
(c)	*State **one** way micro-organisms contribute to the carbon cycle.* They decay (or decompose) the dead matter **<u>or</u>** other correct
(d)	*Suggest **two** possible ways Ireland can directly reduce carbon dioxide emissions.* Plant more trees / reduce the herd size / plant more clover / introduce laws/ stop using fossil fuels / other correct ***Any two***
(e)	*Name **one** other nutrient recycling pathway you have studied.* Nitrogen

Question 4	2(5) + 5(2)
(a)	*Name the cell organelle shown in the image.* Chloroplast
(b)	*Name the green pigment required for photosynthesis.* Chlorophyll
(c)	*In which group of organisms would you find the organelle you named in part (a)?* Eukaryotic *Give a reason for your answer.* It is membrane bound
(d)	*Write a balanced chemical equation for photosynthesis.* $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ ***(one point for correct formulas + one point for correct balancing)***
(e)	*Name **one** factor that affects the rate of photosynthesis.* Carbon dioxide concentration **<u>or</u>** light intensity **<u>or</u>** other correct

Question 5	2(5) + 5(2)
(a)	*Which stage of mitosis is shown in the diagram?* Metaphase
(b)	*Describe **one** event that occurs during this stage.* (Pairs of) chromosomes line up along equator **<u>or</u>** (each pair of) chromosomes is attached to spindle fibres
(c)	*How many daughter cells will result from mitosis of this cell?* 2
(d)	*What is the diploid number of the cell in the diagram?* 4
(e)	***In the box provided above**, sketch a diagram to show the next stage of mitosis.* Sketch showing chromosomes being pulled to the poles
(f)	*Give **one** function of mitosis in multi-cellular organisms.* Growth **<u>or</u>** repair **<u>or</u>** other correct
(g)	*What term describes uncontrolled mitosis and cell division?* Cancer

Question 6		**2(5) + 5(2)**
(a)	*Give **one** structural feature of the stem in the photograph that identifies it as dicotyledonous.* Vascular bundles are arranged in a ring	
(b)	***On the diagram above**, draw an arrow from the letter **A** to the ground tissue of the stem.* Correct arrow drawn to ground tissue	
(c)	*Name these **two** vascular tissues **and** give **one** function of **each**.* *Names:* Xylem Phloem *Function of xylem:* Transport water and minerals *Function of phloem:* Transport food	
(d)	*Name the corresponding structures on stems.* Lenticels	

Question 7		**2(5) + 4(2) + 2(1)**
(a)	*Name any **one** region of the vertebral column.* Cervical **<u>or</u>** thoracic **<u>or</u>** lumbar **<u>or</u>** sacral **<u>or</u>** coccyx	
(b)	***On the diagram**, mark the region of the vertebral column you named in part (a) above with the letter **X**.* 'X' indicating correct region	***[must match part (a)]***
(c)	*How many vertebrae are in the region you named at part (a) above?* Cervical: 7 **<u>or</u>** thoracic: 12 **<u>or</u>** lumbar: 5 **<u>or</u>** sacral: 5 **<u>or</u>** coccyx: 4	***[must match part (a)]***
(d)	*Give **one** function of these discs.* Shock absorption **<u>or</u>** friction-free movement **<u>or</u>** articulation	
(e)	***On the diagram, draw an arrow** from the letter **D** to the location of **one** of the discs of cartilage.* D indicating a vertebral disc	
(f)	*What type of joint is found between vertebrae?* Slightly movable	
(g)	*Write one of these disorders down and give **one** cause **and one** treatment for the named disorder.* *Arthritis:* *Cause:* Genetic **<u>or</u>** wear and tear **<u>or</u>** other correct *Treatment:* Physiotherapy **<u>or</u>** other correct **<u>OR</u>** *Osteoporosis:* *Cause:* Genetic **<u>or</u>** hormonal **<u>or</u>** other correct *Treatment:* Physiotherapy **<u>or</u>** other correct	

Section B	**Best 2**	**60**

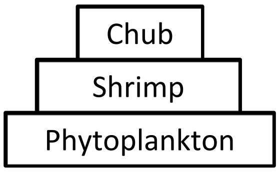
Question 8				**30**
(a)	*Explain the terms biotic **and** abiotic as used in ecology.*			
	Biotic: Living factor			**3**
	Abiotic: Non-living factor			**3**
(b)	(i)	*Describe how the frequency data would have been obtained.*		
		Quadrat / random (sample) / how random / count (plants) / repeat **or** a number of times / a correct calculation described	***Any three***	**3(3)**
	(ii)	*Calculate the percentage frequency for both grass **and** clover when the herbivore was present.*		
		Grass: 60%		**3**
		Clover: 40%		**3**
	(iii)	*Suggest **one** possible conclusion that can be made from the data.*		
		Herbivores reduce grass or clover cover **or** other correct		**3**
	(iv)	*Suggest **one** possible source of error in this investigation **and** how this source of error may be minimised.*		
		Source of error: Source of error identified		**3**
		How minimised: Correct description of how to minimise error		**3**

Question 9				**30**
(a)	(i)	*Name **one** factor, other than temperature, that affects the rate of activity of an enzyme.*		
		pH		**3**
	(ii)	*Explain the term optimum activity in relation to enzymes.*		
		Working at its maximum rate		**3**
(b)	(i)	*Name the enzyme you used **and** its substrate(s) **and** product(s).*		
		Named enzyme		**3**
		Named substrate(s)	***(must match named enzyme)***	**3**
		Named product(s)	***(must match named enzyme)***	**3**
	(ii)	*Describe how you carried out the investigation.*		
		Boil enzyme (or other denaturation method) / unboiled (or fresh) enzyme / named vessel / pH buffer / water bath at 25°C / left for a time / measure activity / how measured	***Any three***	**3(3)**
	(iii)	*Describe the results of this investigation.*		
		Boiled enzyme showed no activity		**3**
		Fresh enzyme showed activity		**3**

Question 10			**30**
(a)		*Give **two** factors, other than exercise, that have an effect on the circulatory system.* Diet / smoking / other correct ***Any two***	**2(3)**
(b)	(i)	*Describe how the student measured resting PR **or** resting BR.* *PR:* Counted number of pulses per minute **<u>OR</u>** *BR:* Counted number of breaths per minute	**3** **3**
	(ii)	*In the space provided, draw (using pencil) a suitable graph to represent the data of **either** pulse rate (PR) **or** breathing rate (BR).* PR or BR on the y-axis <u>**and**</u> Resting, Moderate exercise and Intense exercise on the x-axis Bar chart <u>**or**</u> line graph <u>**or**</u> dot plot <u>**or**</u> other suitable graph drawn	 **3** **3**
	(iii)	*State what would happen to the PR **or** BR after exercise has stopped.* Decrease	**3**
	(iv)	*Suggest a reason for the student repeating the investigation three times.* More accurate data	**3**
	(v)	*During this activity, body temperature also increased. Suggest a reason for this.* Respiration (or energy production) increased releasing heat <u>**or**</u> other correct	**3**
	(vi)	*Describe a safety precaution the student would have taken in carrying out this investigation.* Valid safety precaution described	**3**

Section C	Best 4	4(60)

Question 11			60
(a)	(i)	Name the component parts **X and Y**.	
		X: Phosphate	**3**
		Y: Sugar or deoxyribose	**3**
	(ii)	*Name the type of bonding (**Z**) that links the base pairs in the DNA molecule.*	
		Hydrogen (or H)	**3**
(b)	(i)	*Explain the underlined term.*	
		Form of a gene	**3**
	(ii)	*Using the letters described in the paragraph, write down the genotypes of the parents.*	
		XNY–	**3**
		XNXn	**3**
	(iii)	*Write down the genotypes of all **four** children.*	
		1. XNXN	**3**
		2. XNXn	**3**
		3. XNY–	**3**
		4. XnY–	**3**
	(iv)	*Explain why males are more likely than females to be red-green colour-blind.*	
		Males only have one allele (for red-green colour vision)	**3**
	(v)	*Name **one** other common sex-linked trait.*	
		Haemophilia **or** other correct	**3**
(c)	(i)	*What term is used to describe the state of non-division of a cell?*	
		Interphase	**3**
	(ii)	*Draw a labelled diagram (e.g. a pie chart) to illustrate the relative lengths of time of non-division and division during the cell cycle.*	
		Diagram: Showing one phase as a long period of time **and** another phase as a short period of time	**1**
		Labels: Interphase labelled as part of the long period	**1**
		Mitosis labelled as part of the short period	**1**
	(iii)	*Describe the main events of DNA replication.*	
		DNA unwinds / strands separate / corresponding nucleotides added in / two new chromosomes produced ***Any three***	**3(3)**
	(iv)	*Is DNA replication an anabolic or a catabolic process? Explain your answer.*	
		Anabolic **and** because a larger molecule is produced from smaller molecules.	**3**
	(v)	*Mitosis is one type of division. Name **and** briefly describe the other type.*	
		Name: Meiosis	**3**
		Description: Division that produces four daughter cells **or** division where the number of chromosomes is halved	**3**

Question 12			**60**
(a)	(i)	*Explain the term pollution.*	
		Harmful addition to the environment	**3**
	(ii)	*Describe **one** effect of a **named** pollutant **and one** way it may be controlled.*	
		Effect described	**3**
		Control described	**3**
(b)	(i)	*What other term in the passage is used to describe 'invasive alien species'?*	
		Non-native (species)	**3**
	(ii)	*Give **two** examples of aquatic environmental factors.*	
		Light intensity / temperature / pH / salinity / other correct ***Any two***	**2(3)**
	(iii)	*Define the term conservation **and** describe **one** part of the passage that is promoting conservation.*	
		Management of the environment	**3**
		Awareness (or implementation) of biosecurity measures	**3**
	(iv)	*Suggest **one** effect of not implementing conservation measures in Ireland.*	
		Species population decline **or** species become extinct **or** less biodiversity **or** other correct	**3**
	(v)	*Using the organisms named in the passage, write down a food chain containing three trophic levels.*	
		Phytoplankton → Shrimp → European Chub	**3**
	(vi)	*Draw a pyramid of numbers for the food chain you described above.*	
			
		Correct order with Seaweed on the bottom	**3**
		Upright pyramid drawn	**3**
(c)	(i)	*Explain the term habitat.*	
		Place where an organism lives	**3**
	(ii)	*In relation to the herbivores, suggest **two** reasons for their fluctuating populations as shown in each graph.*	
		Predator numbers change / disease / migration / food levels change / other correct ***Any two***	**2(3)**
	(iii)	*Estimate the mean maximum population of lion over the first 20 years.*	
		2 – 3 thousand	**3**
	(iv)	*Suggest a reason for the population of lions in the habitat being lower than the population of their prey.*	
		Energy is lost between trophic levels **or** other correct	**3**
	(v)	***Copy the middle graph into your answerbook** and continue the graph to illustrate the population of lions over the next 10 years based on the corresponding numbers of their prey.*	
		Graph line showing an increase	**3**
	(vi)	*Explain why wildebeest and zebra are in competition with each other in this habitat.*	
		Same food source **or** same habitat **or** other correct	**3**
	(vii)	*Name **one** other ecological relationship.*	
		Parasitism **or** symbiosis	**3**

Question 13			**60**
(a)	(i)	*Explain the term obligate parasite.*	
		Needs a host cell	**3**
		To reproduce	**3**
	(ii)	*Give **one** way in which viruses are economically important to humans.*	
		Used to make vaccines **or** used in genetic engineering **or** other correct	**3**
(b)	(i)	*Distinguish between the terms autotrophic **and** heterotrophic by writing a brief sentence on each.*	
		Autotrophic: (Organism) that makes its own food	**3**
		Heterotrophic: (Organism) that obtains its food from other organisms	**3**
	(ii)	*Give **one** example of an autotrophic bacterium **and one** example of a heterotrophic bacterium.*	
		Named autotrophic bacterium	**3**
		Named heterotrophic bacterium	**3**
	(iii)	*State **two** factors that affect the growth of bacteria.*	
		pH / temperature / other correct ***Any two***	**2(3)**
	(iv)	*What is an antibiotic?*	
		Chemical that kills bacteria	**3**
	(v)	*Describe how antibiotic resistance develops in bacterial populations.*	
		Competition between sensitive and resistant strains **or** explained	**3**
	(vi)	*Explain why antibiotics are not effective against the common cold.*	
		The common cold is a virus and antibiotics have no effect on viruses.	**3**
(c)	(i)	*Name the structures labelled **A, B** and **C**.*	
		A: *Cell wall	**1**
		B: *Nucleus	**1**
		C: *Food vacuole	**1**
	(ii)	*To which kingdom does yeast belong?*	
		*Fungi	**3**
	(iii)	*Name another organism that belongs to the kingdom you named at part (c) (ii) above.*	
		Rhizopus **or** other correct	**3**
	(iv)	*What name is given to asexual reproduction in yeast cells?*	
		*Budding	**3**
	(v)	*Describe in detail the process of asexual reproduction in yeast cells.*	
		Organelles (or nucleus) divide / new bud forms / organelles (or nucleus) move into bud / new cell detaches (or can remain attached) / colony forms ***Any three***	**3(3)**
	(vi)	*Give **one** way yeast cells differ from bacterial cells.*	
		They are eukaryotic **or** they contain organelles **or** they are larger **or** other correct	**3**

Question 14			**60**
(a)	(i)	*What is meant by the term infertility?*	
		Inability to produce offspring	**3**
	(ii)	*Choosing either fibroids **or** endometriosis, give **one** cause **and one** treatment.*	
		Fibroids: / **OR** / *Endometriosis:*	
		Cause: Genetic **or** hormonal **or** other correct **OR** Genetic **or** hormonal **or** other correct	**3**
		Treatment: Surgery **or** other correct **OR** Surgery **or** other correct	**3**
(b)	(i)	*Name the parts labelled **A, B** and **C**.*	
		A: *Prostate	**2**
		B: *Epididymis	**2**
		C: *Penis	**2**
	(ii)	*Give a function for the parts labelled **B and C**.*	
		Function of B: Store sperm **or** mature sperm	**3**
		Function of C: Sexual intercourse	**3**
	(iii)	*Name the male sex hormone produced by the part labelled **X**.*	
		*Testosterone	**3**
	(iv)	*Draw **and** label the structure of a sperm cell.*	
		Drawing: Head **and** midpiece **and** tail	**3**
		Labels: Head / nucleus / midpiece (or neck) / mitochondria / tail ***Any three***	**3(1)**
	(v)	*State the approximate survival time of sperm within the female reproductive tract.*	
		0 – 7 days	**3**
	(vi)	*Name the structure through which both urine and sperm travel.*	
		*Urethra	**3**
(c)	(i)	*Draw a labelled diagram of the human female reproductive system.*	
		Diagram: Vagina **and** uterus **and** fallopian tube **and** ovary ***(any one missing = 3 marks)***	**6**
		Labels: Vagina / vulva / cervix / uterus / endometrium / fallopian tube / ovary ***Any six***	**6(1)**
		***On your labelled diagram**, indicate where each of the following occur:*	
		1. Egg production: Ovary indicated	**2**
		2. Fertilisation: Fallopian tube indicated	**2**
		3. Implantation: Uterus (or endometrium) indicated	**2**
	(ii)	*Give the names of **two** structures that are formed in the days following fertilisation…*	
		*Morula	**3**
		*Blastocyst	**3**

Question 15				60
(a)	(i)	*Distinguish between aerobic respiration **and** anaerobic respiration, by writing a brief sentence on each.*		
		Aerobic:	Releases a large amount of energy **or** uses oxygen	3
		Anaerobic:	Releases a small amount of energy **or** does not require oxygen	3
	(ii)	*Identify the organelle shown in the T.E.M. image that is responsible for aerobic respiration.*		
		Mitochondrion		3
(b)	(i)	*Name stage 1 **and** state where in a cell it occurs.*		
		*Glycolysis		2
		*Cytosol		2
	(ii)	*How many carbon atoms are present in pyruvate?*		
		*3		2
	(iii)	*What name is given to this new molecule?*		
		*Acetyl (co-enzyme A)		3
	(iv)	*Comment on the relative amounts of ATP produced by both of these stages.*		
		Stage 1 produces a small amount of ATP **and** stage 2 produces a large amount of ATP		3
	(v)	*Name the parts labelled **1**, **2** and **3**.*		
		1.	*Adenine	2
		2.	*Ribose	2
		3.	*Phosphates	2
	(vi)	*What does NAD stand for?*		
		Nicotinamide adenine dinucleotide		3
	(vii)	*Describe, in detail, the roles NAD **and** oxygen play in aerobic respiration.*		
		NAD:	Carries high energy electrons or protons	3
		Oxygen:	receives electrons or protons **or** forms water	3
(c)	(i)	*What is an enzyme?*		
		Protein catalyst		3
	(ii)	*Name **one** anabolic **and one** catabolic enzyme you have studied.*		
		Named anabolic enzyme		3
		Named catabolic enzyme		3
	(iii)	*What is meant by the term specificity in relation to enzyme function?*		
		Enzyme will act on only one substrate		3
	(iv)	*1. What is an immobilised enzyme?*		
		An enzyme that has been trapped in an insoluble material (or attached to each other)		3
		*2. Give **one** advantage of immobilised enzymes.*		
		Pure product formed **or** more stable than free enzyme **or** reusable		3
		3. Describe how enzymes may be immobilised.		
		Enzyme added to (sodium) alginate / calcium chloride added to solidify gel		2(3)

Question 16	**Any two of (a), (b), (c), (d)**	**30, 30**

Question 16 (a)			**30**
(i)	*Name the parts labelled **X, Y and Z**.*		
	X: Style		**1**
	Y: Ovary		**1**
	Z: Ovule		**1**
(ii)	*Give **one** other way in which pollen can be carried to the stigma of a flower.*		
	Wind <u>**or**</u> water		**3**
(iii)	***Copy the diagram and** draw in a pollen tube to show the path taken by the male gametes.*		
	Carpel drawn with the correct path taken by the pollen tube		**3**
(iv)	*Describe how this nucleus gives rise to the male gametes.*		
	(Generative nucleus) divides by mitosis		**3**
(v)	*Give an account of the development of the egg cell within structure **Z**.*		
	Diploid / megaspore mother cell / divides by meiosis / to produce four haploid cells (or tetrad) / three degenerate / divides by mitosis 3 times / embryo sac containing 8 nuclei (cells) / one becomes egg (cell) / two polar nuclei (formed)	***Any four***	**4(3)**
(vi)	*Describe what happens during **each** fertilisation.*		
	Egg cell fuses with one sperm nucleus to form a diploid zygote		**3**
	Two polar nuclei fuse with the (other) sperm nucleus to form triploid endosperm		**3**

Question 16 (b)			**30**
(i)	*Name the parts labelled **A, B and C**.*		
	A: Semi-circular canals		**1**
	B: Cochlea		**1**
	C: Eardrum		**1**
(ii)	*Give the functions of the parts labelled **A and B**.*		
	Function of A: Balance		**3**
	Function of B: Hearing		**3**
(iii)	*What is the name of the bone that helps to protect the internal parts of the ear?*		
	Skull		**3**
(iv)	*What is the collective terms for these three bones?*		
	Ossicles		**3**
(v)	*The Eustachian tube connects the middle ear to another structure. Name this other structure.*		
	Throat or pharynx		**3**
(vi)	*Name any **two** sense organs, other than the eye and the ear.*		
	Skin / nose / tongue	***Any two***	**2(3)**
(vii)	*Name **one** disorder of the eye **or** the ear **and** suggest **one** treatment.*		
	Named disorder		**3**
	Treatment	***(must match named disorder)***	**3**

Question 16 (c)		**30**
(i)	*State the precise location in the body where blood cells are made.*	
	*Red bone marrow	**3**
(ii)	*Name the blood cells labelled **X**.*	
	*Red (blood cells)	**3**
(iii)	*Give **one** function of the cells labelled **X and** give **one** adaptation that enables them to carry out this role.*	
	Function: Carry oxygen	**3**
	Adaptation: Possess haemoglobin **or** no nucleus **or** biconcave shape **or** very small cells **or** are flexible	**3**
(iv)	Distinguish between active immunity **and** passive immunity by writing a brief sentence on each.	
	Active immunity: Antibodies are made by the organism itself	**3**
	Passive immunity: Antibodies are received that have been made in another organism	**3**
(v)	*1. What is an antigen?*	
	A chemical that stimulates the production of antibodies	**3**
	*2. Name any **three** types of T cell.*	
	Helper / killer / suppressor (or regulatory) / memory ***Any three***	**3(3)**

Question 16 (d)		**30**
(i)	*Is the seed in diagram **A** classified as monocotyledonous or dicotyledonous? Give **one** reason for your answer.*	
	Monocotyledonous **and** correct reason	**3**
(ii)	***Copy either seed into your answerbook and** label the following parts: **testa**; **cotyledon**; **embryo***	
	Each part labelled correctly	**3(1)**
(iii)	*Name **one** example of a monocotyledonous seed **and one** example of a dicotyledonous seed.*	
	Named monocot seed	**3**
	Named dicot seed	**3**
(iv)	*Which part of the embryo in a germinating seed gives rise to each of the following parts of the seedling?*	
	1. The root: *Radicle	**3**
	2. The shoot: *Plumule	**3**
(v)	*1. What is meant by the term dispersal?*	
	The scattering or spreading of the seeds away from the parent plant	**3**
	*2. Name **two** methods used by plants to disperse seeds.*	
	Wind / water / animal / self ***Any two***	**2(3)**
	*3. State **one** advantage of dispersal to the plant.*	
	Less competition with the parent plant **or** colonise new habitats **or** other correct	**3**

Question 17	**Any two of (a), (b), (c), (d)**	**30, 30**

Question 17 (a)		
(i)	*Explain the term homeostasis.* Maintenance of a constant internal environment	**3**
(ii)	*Name an excretory product common to all three organs involved in excretion.* *Water	**3**
(iii)	*1. Draw a labelled diagram of the nephron, including its blood supply.* *Diagram:* Bowman's capsule **and** glomerulus **and** proximal convoluted tubule (or PCT) **and** loop of Henle **and** distal convoluted tubule (or DCT) **and** collecting duct **and** blood supply ***(Any one missing = 3)***	**6**
	Labels: Bowman's capsule / glomerulus / PCT / loop of Henle / DCT / collecting duct / blood supply ***Any three***	**3(1)**
	*2. **On your diagram**, place the letter **X** on the location where filtration occurs **and** place the letter **Y** on the location where reabsorption occurs.* X on Bowman's capsule **or** glomerulus	**3**
	Y on PCT **or** loop of Henle **or** DCT **or** collecting duct	**3**
(iv)	*What might cause this increase?* Lack of water in the bloodstream **or** not drinking enough water **or** (intense) exercise **or** other correct	**3**
(v)	*Where in the nephron does ADH act?* Distal convoluted tubule **or** collecting duct	**3**
(vi)	*What is the effect on the nephron of increased ADH levels?* More water is reabsorbed (from the nephron into the bloodstream) **or** makes the DCT or collecting duct more permeable to water	**3**

Question 17 (b)				**30**
(i)	*Name **one** other part of the central nervous system.* *Spinal cord			**3**
(ii)	*1. What is the name of this tissue?* *Meninges			**3**
	2. How many layers are present in this tissue? *Three			**3**
	3. Name an infection associated with this tissue. *Meningitis			**3**
(iii)	*1. Name the parts of the brain labelled **A, B and C**.* *A:* *Medulla oblongata			**1**
	B: *Cerebellum			**1**
	C: *Cerebrum			**1**
	*2. Give **one** function of **each** labelled part.* *Function of A:* Breathing **or** involuntary actions **or** other correct			**3**
	Function of B: Muscular coordination and balance			**3**
	Function of C: Memory **or** intelligence **or** emotion **or** speech **or** hearing **or** other correct			**3**
(iv)	*In relation to paralysis or Parkinson's disease, give **one** possible cause **and one** possible treatment.*			
		Paralysis		*Parkinson's disease*	
	Cause:	Injury **or** other correct	**OR**	Genetic **or** other correct	**3**
	Treatment:	Physiotherapy **or** other correct		Administration of L-dopa **or** other correct	**3**

Question 17 (c)		**30**
(i)	*Name the parts labelled **A, B, C, D, E and F**.*	
	A: *Aorta	**1**
	B: *Left atrium	**1**
	C: *Bicuspid valve	**1**
	D: *Septum	**1**
	E: *Semilunar valve	**1**
	F: *Vena cava	**1**
(ii)	*State **one** reason why this is important.*	
	To prevent deoxygenated blood mixing with oxygenated blood	**3**
(iii)	*Name this type of muscle **and** give **one** characteristic of this specialised muscle tissue.*	
	Name: *Cardiac	**3**
	Characteristic: Slow to tire	**3**
(iv)	*The wall of the left ventricle is thicker than the wall of the right ventricle. Explain why this is necessary.*	
	The left side of the heart pumps blood around the systemic circuit (or around the body)	**3**
(v)	*State their respective locations in the heart **and** describe how each carries out their role in heartbeat control.*	
	Location of SA node: wall of right atrium	**3**
	Role of SA node: sends out (electrical) impulses causing the atria to contract	**3**
	Location of AV node: wall between the ventricles or between septum	**3**
	Role of AV node: sends out (electrical) impulses causing the ventricles to contract	**3**

Question 17 (d)		**30**
(i)	*Distinguish between mechanical **and** chemical digestion by writing a brief sentence on **each**.*	
	Mechanical: breakdown of food <u>physically</u>	**3**
	Chemical: breakdown of food using <u>enzymes</u> (or acid)	**3**
(ii)	*Digestion and absorption are two of the four stages of nutrition. Name the other **two** stages.*	
	*Ingestion	**3**
	*Egestion	**3**
(iii)	*Name the parts labelled **A, B, C, D, E and F**.*	
	A: *Oesophagus	**1**
	B: *Liver	**1**
	C: *Gall bladder	**1**
	D: *Mouth	**1**
	E: *Stomach	**1**
	F: *Large intestine	**1**
(iv)	*Give **two** functions of the structure labelled **B**.*	
	Make bile / detoxify / deamination of amino acids / storage of vitamins / heat production / breaks down red blood cells / other correct ***Any two***	**2(3)**
(v)	*State **one** benefit of having fibre in the diet.*	
	Aids peristalsis **<u>or</u>** prevents constipation **<u>or</u>** other correct	**3**
(vi)	*Give **one** benefit of bacteria living in the digestive tract.*	
	Out-compete harmful bacteria **<u>or</u>** digest cellulose **<u>or</u>** make vitamins **<u>or</u>** other correct	**3**

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